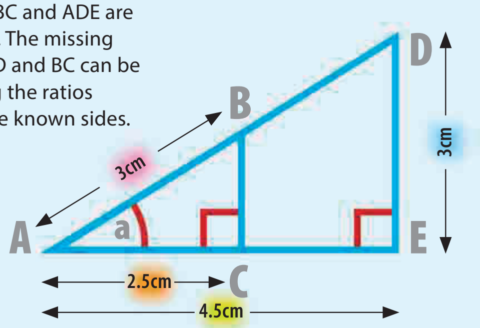
MISSING SIDES IN SIMILAR TRIANGLES

The proportional relationships between the sides of similar triangles can be used to find the value of sides that are missing, if the lengths of some of the sides are known.

▷ **Similar triangles**
Triangles ABC and ADE are similar (AA). The missing values of AD and BC can be found using the ratios between the known sides.



Finding the length of AD

To find the length of AD, use the ratio between AD and its corresponding side AB, and the ratio between a pair of sides where both the lengths are known – AE and AC.

Write out the ratios between the two pairs of sides, each with the longer side above the shorter side. These ratios are equal.

$$\frac{AD}{AB} = \frac{AE}{AC}$$

Substitute the values that are known into the ratios. The numbers can now be rearranged to find the length of AD.

AD is the unknown

$$\frac{AD}{3} = \frac{4.5}{2.5}$$

Rearrange the equation to isolate AD. In this example this is done by multiplying both sides of the equation by 3.

multiply both sides by 3

$$AD = 3 \times \frac{4.5}{2.5}$$

multiply by 3 to isolate AD

Do the multiplication to find the answer, and add the units to the answer that has been found. This is the length of AD.

$$AD = 5.4\text{cm}$$

Finding the length of BC

To find the length of BC, use the ratio between BC and its corresponding side DE, and the ratio between a pair of sides where both the lengths are known – AE and AC.

Write out the ratios between the two pairs of sides, each with the longer side above the shorter side. These ratios are equal.

$$\frac{DE}{BC} = \frac{AE}{AC}$$

Substitute the values that are known into the ratios. The numbers can now be rearranged to find the length of BC.

$$\frac{3}{BC} = \frac{4.5}{2.5}$$

Rearrange the equation to isolate BC. This may take more than one step. First multiply both sides of the equation by BC.

multiply both sides by BC

multiply both sides by BC

$$3 = \frac{4.5}{2.5} \times BC$$

Then rearrange the equation again. This time multipy both sides of the equation by 2.5.

multiply both sides by 2.5

$$3 \times 2.5 = 4.5 \times BC$$

multiply both sides by 2.5

BC can now be isolated by rearranging the equation one more time – divide both sides of the equation by 4.5.

divide both sides by 4.5

$$BC = \frac{3 \times 2.5}{4.5}$$

divide both sides by 4.5

Do the multiplication to find the answer, add the units, and round to a sensible number of decimal places.

1.6666.... is rounded to 2 decimal places

$$BC = 1.67\text{cm}$$